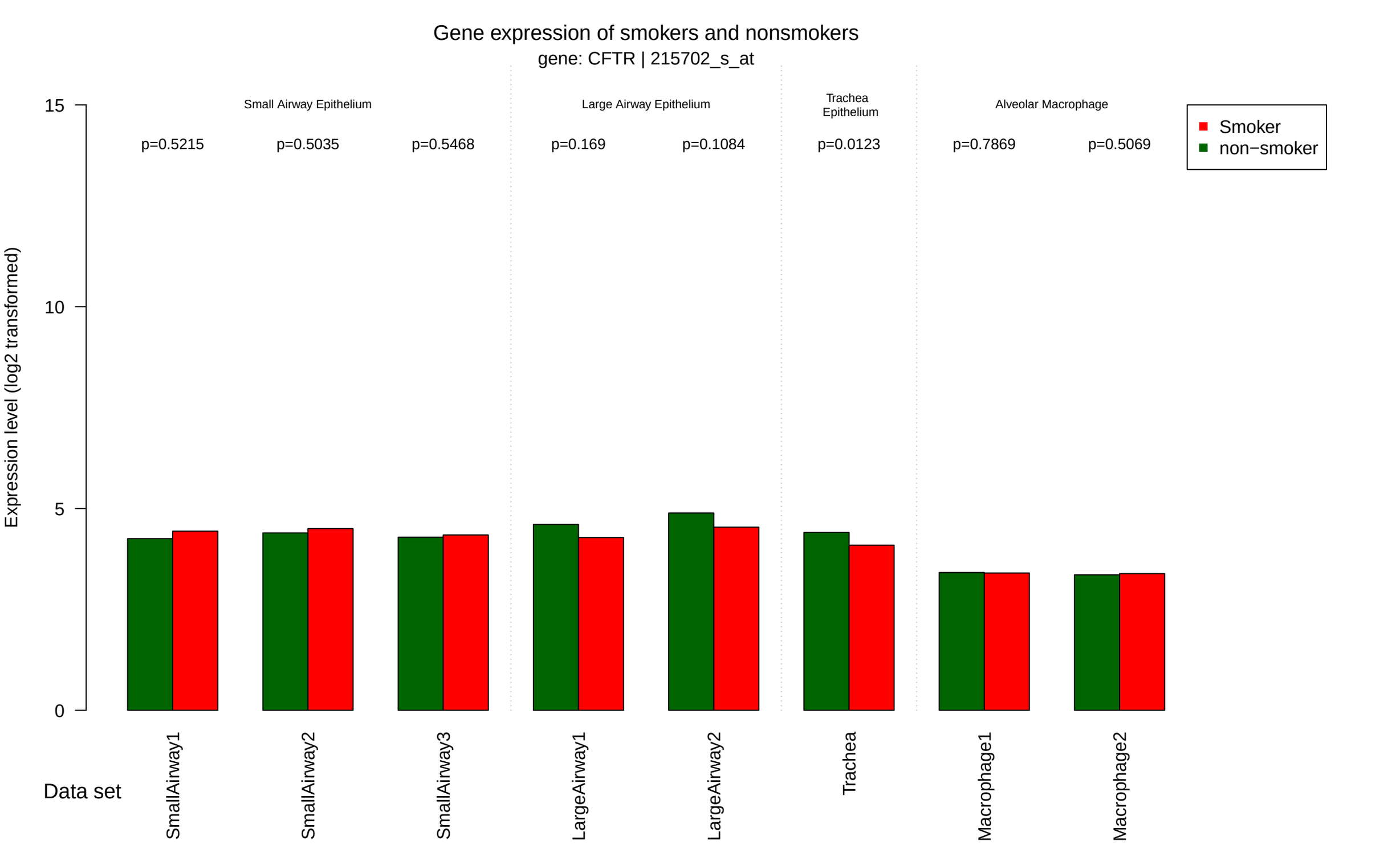
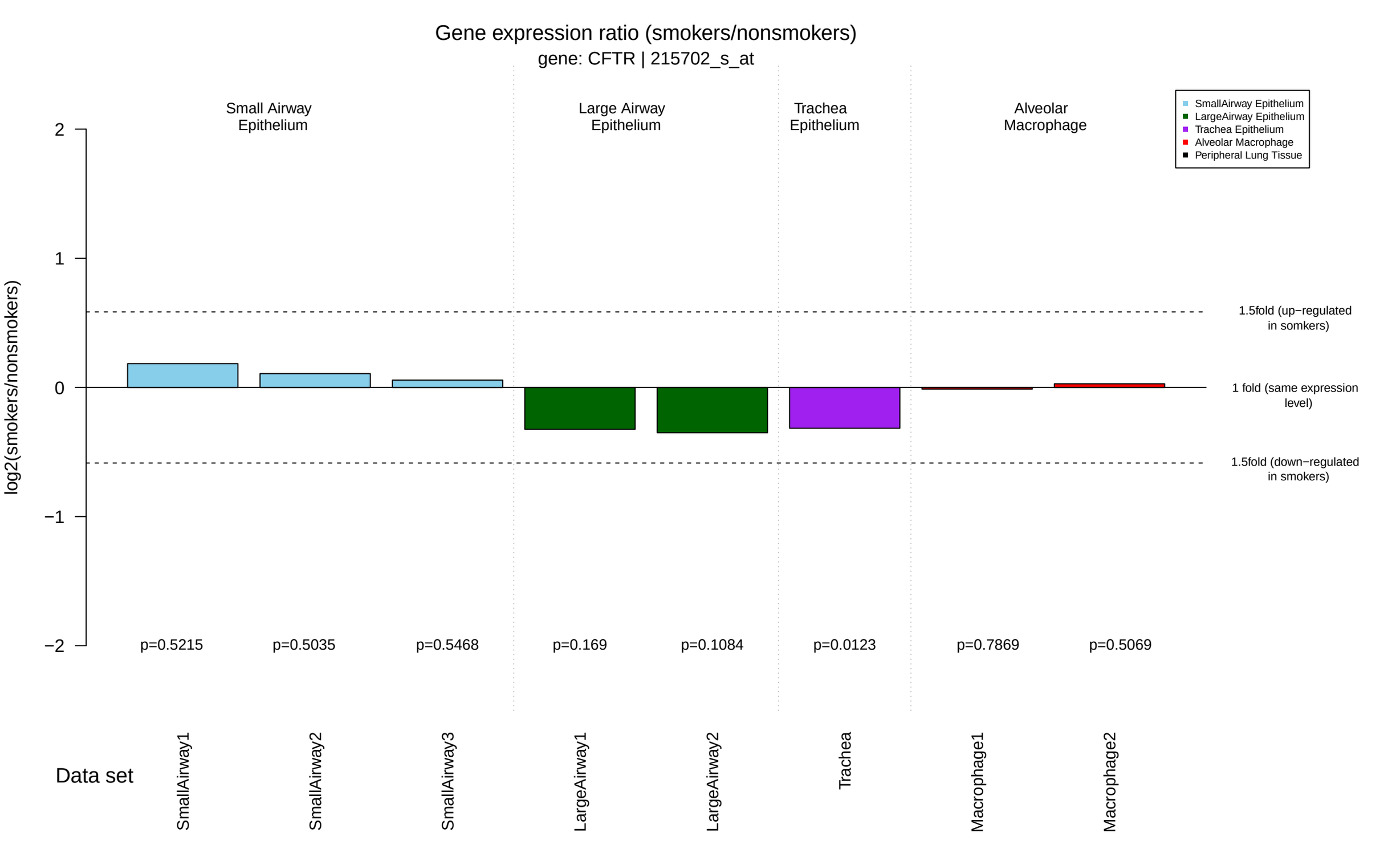
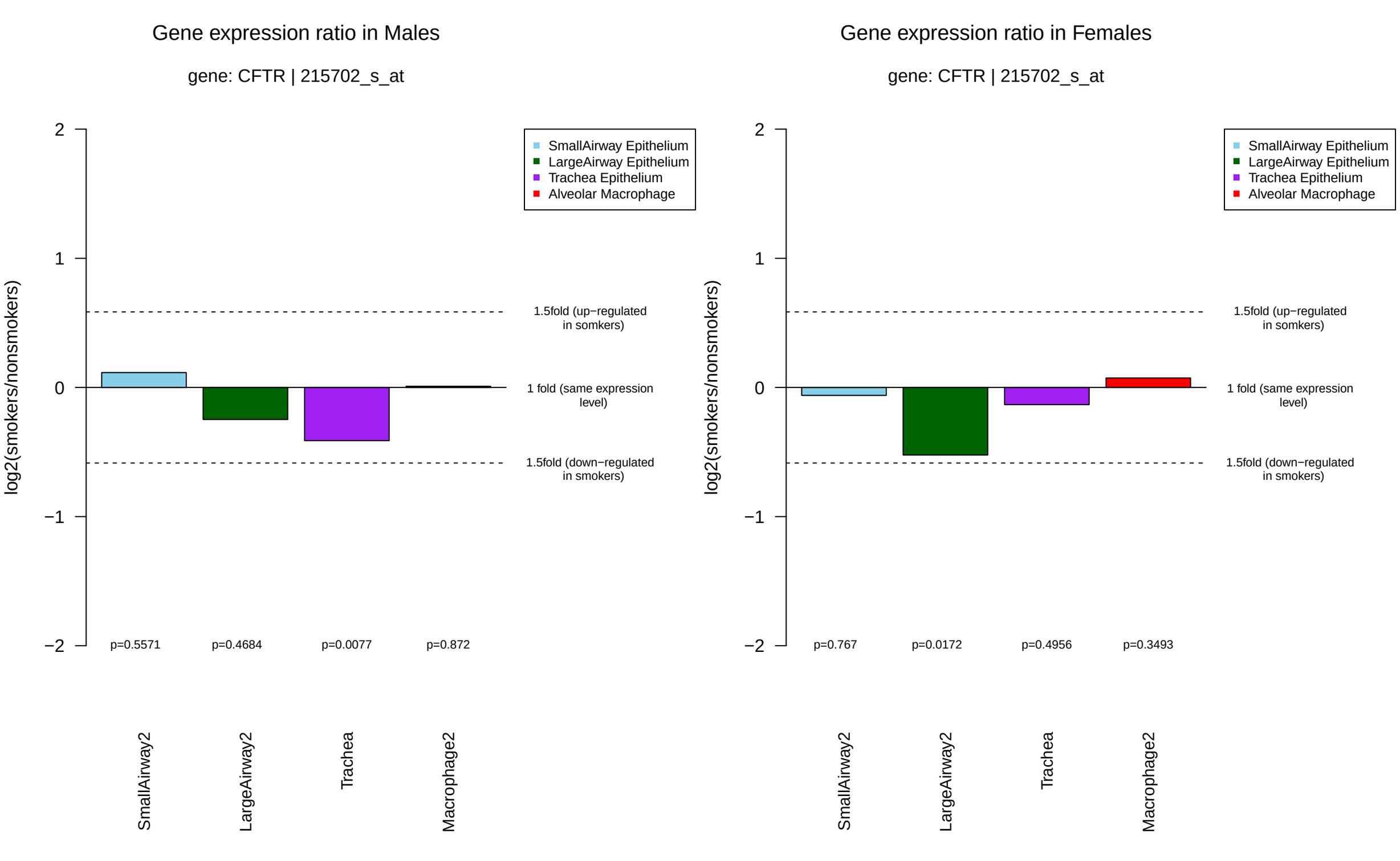
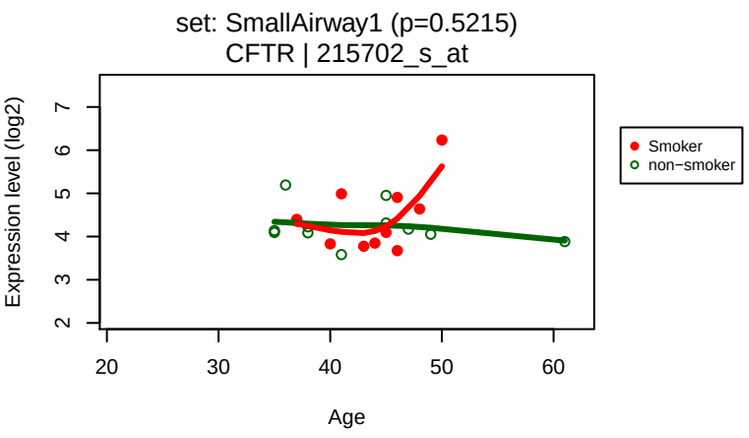
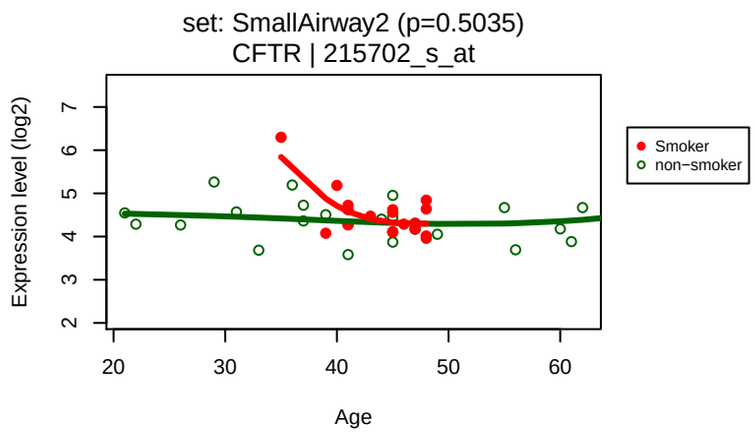
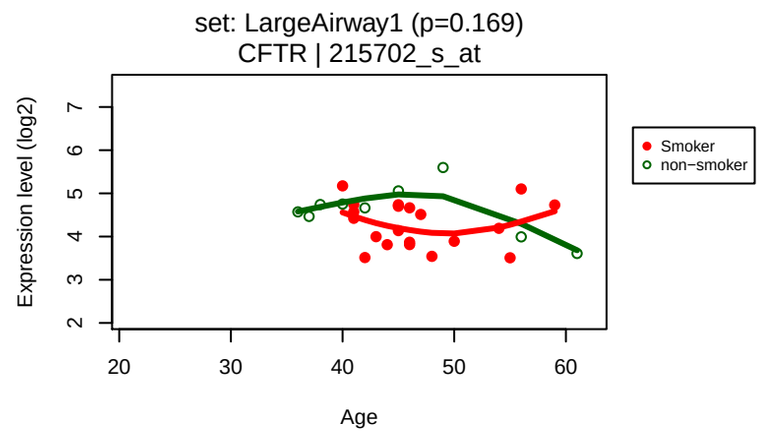

Gene expression of smokers and nonsmokers
gene: CFTR | 215702_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5215
p=0.5035
p=0.5468
p=0.169
p=0.1084
p=0.0123
p=0.7869
p=0.5069
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

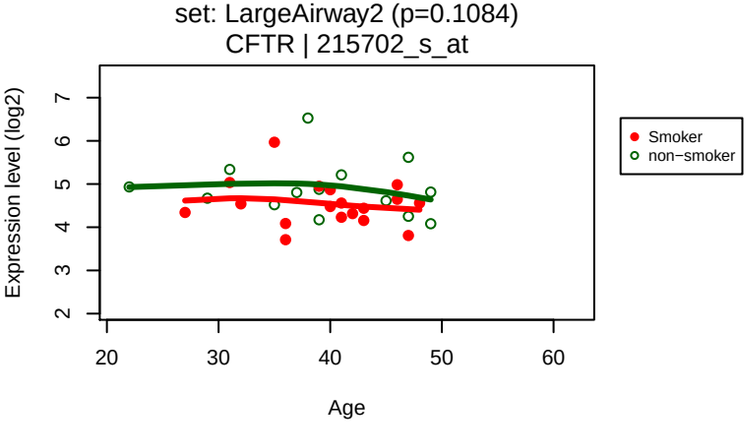

Gene expression ratio (smokers/nonsmokers)
gene: CFTR | 215702_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5215
p=0.5035
p=0.5468
p=0.169
p=0.1084
p=0.0123
p=0.7869
p=0.5069
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

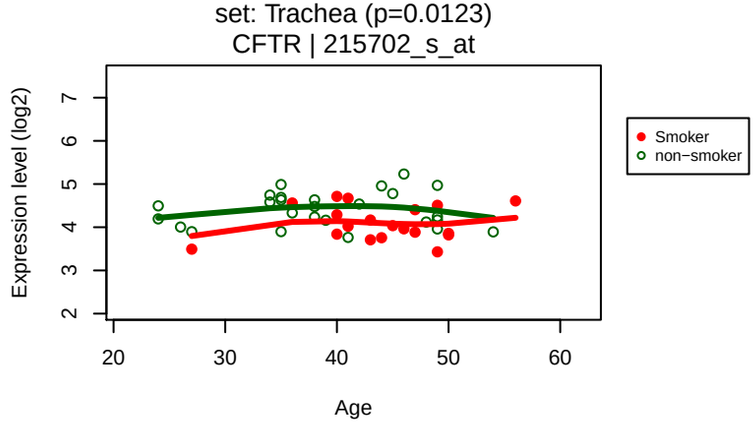

Gene expression ratio in Males
gene: CFTR | 215702_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5571
p=0.4684
p=0.0077
p=0.872
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CFTR | 215702_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.767
p=0.0172
p=0.4956
p=0.3493
SmallAirway2
LargeAirway2
Trachea
Macrophage2

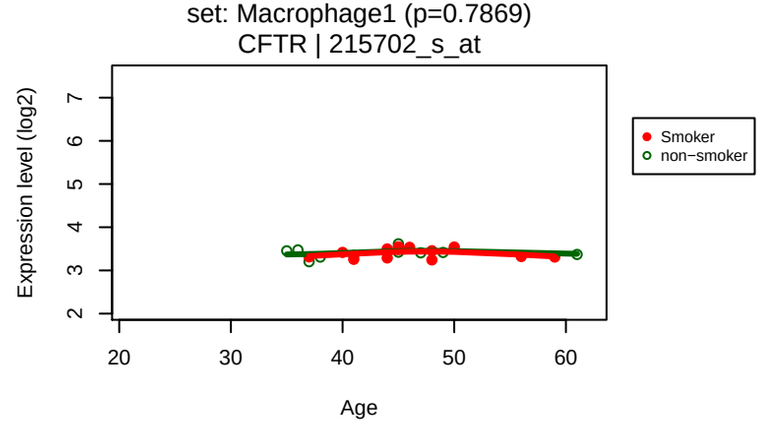

set: SmallAirway1 (p=0.5215)
CFTR | 215702_s_at
Expression level (log2)
Age
Smoker
non−smoker

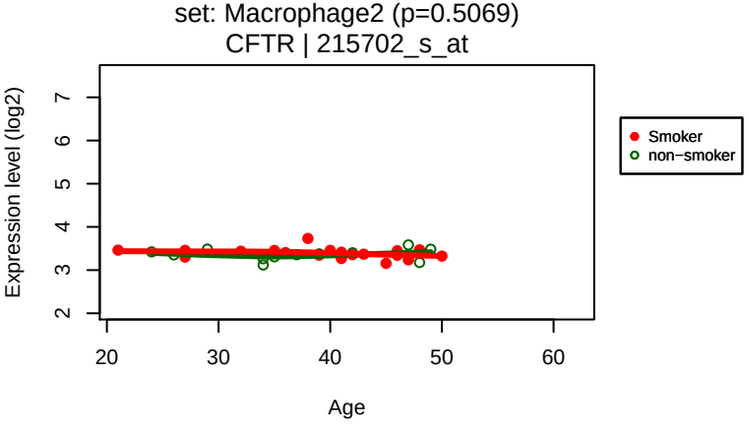

set: SmallAirway2 (p=0.5035)
CFTR | 215702_s_at
Expression level (log2)
Age
Smoker
non−smoker

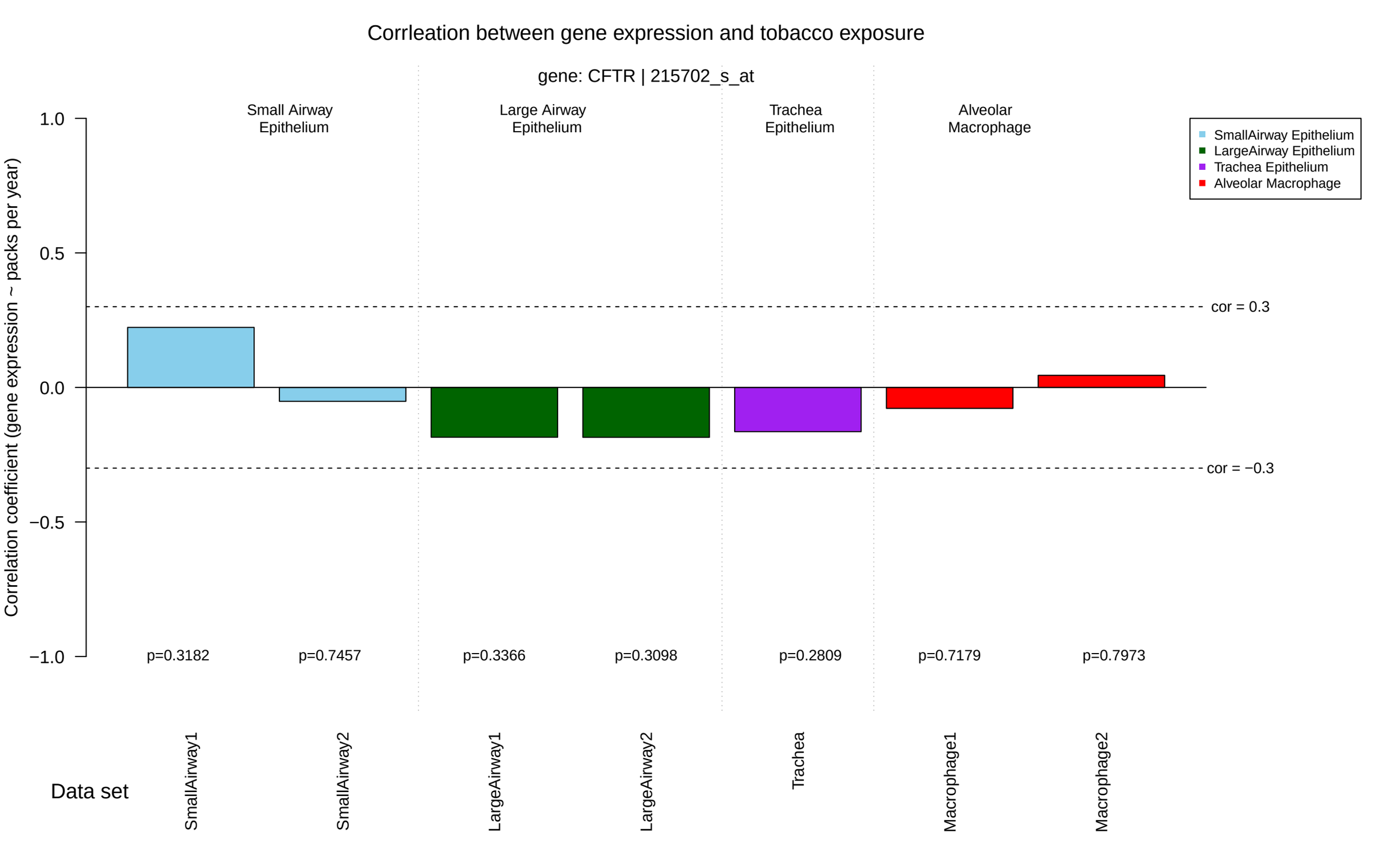

set: LargeAirway1 (p=0.169)
CFTR | 215702_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1084)
CFTR | 215702_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0123)
CFTR | 215702_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7869)
CFTR | 215702_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5069)
CFTR | 215702_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CFTR | 215702_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3182
p=0.7457
p=0.3366
p=0.3098
p=0.2809
p=0.7179
p=0.7973
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2